



Koneru Lakshmaiah Education Foundation

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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026-27, Term-I

Project Problem Statement Submission Form

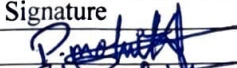
Section / Team Details

Section No.: 10

Team No.: 1

Project Title: PRIORITY SCHEDULING

Team Members

Roll Number	Student Name	Signature
2520080059	P. MOHIT KARTHIK	
2520080049	K. SATYA RISHITH	
2520080016	VARSHITH COMPLE	

1. Abstract

Priority Scheduling is a CPU scheduling technique in which processes are executed based on their assigned priorities. A major problem with priority scheduling is starvation, where low-priority processes may wait for a long time. This project proposes a Linux-based priority scheduling system that demonstrates process management and CPU scheduling using Linux/POSIX system programming concepts. The system creates and manages actual processes, assigns priorities, schedules processes using preemptive and non-preemptive priority scheduling, and uses an aging mechanism to reduce starvation. Linux system calls such as fork(), exec(), waitpid(), kill(), and setpriority() will be used for process creation, control, synchronization, and priority management. The system will measure waiting time, turnaround time, response time, CPU utilization, and context switches. The expected outcome is a command-line based scheduler that provides practical understanding of CPU scheduling, process management, synchronization, signals, IPC, and Linux system programming.

2. Problem Statement

Traditional priority scheduling may cause starvation of low-priority processes when higher-priority processes continuously receive CPU time. The project aims to implement a Linux-based priority scheduler that manages actual processes, supports preemption, and applies aging to improve fairness and reduce starvation while analyzing scheduling performance.

3. Objectives

1. To implement pre-emptive and non pre-emptive Priority Scheduling for Linux processes.
2. To demonstrate process creation, management, synchronization, signals, and Linux/POSIX system calls.
3. To implement an aging mechanism to reduce starvation of low-priority processes.
4. To measure and compare scheduling performance using waiting time, turnaround time, response time, CPU utilization, and context switches.

4. Proposed Methodology

The system will be implemented in C on Linux as a command-line based process scheduler. The scheduler will create worker processes using `fork()` and manage their execution according to assigned priorities. A priority queue will be used to select the next process. Preemption will be handled using Linux signals such as `SIGSTOP` and `SIGCONT`. An aging mechanism will gradually improve the priority of waiting processes to prevent starvation. Process execution and scheduling metrics will be monitored and recorded for performance analysis.

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
CPU Scheduling	Selection and execution of processes based on priority
<code>fork()</code>	Creation of child processes
<code>exec()</code>	Execution of worker programs
<code>waitpid()</code>	Process synchronization and completion monitoring
<code>setpriority()</code>	Assigning/managing process priority
Signals (<code>SIGSTOP</code> , <code>SIGCONT</code> , <code>SIGTERM</code>)	Process suspension, resumption, and termination
Pipes / IPC	Communication between scheduler and processes
Process Management	Creation, execution, monitoring, and termination of processes
Aging	Prevention of starvation of low-priority processes

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520080059	P. MOHIT KARTHIK	Design and implementation of Priority Scheduling algorithm, priority queue, and scheduling logic.
2520080049	K. SATYA RISHITH	Linux process creation, process control, signals, synchronization, and IPC implementation.
2520080016	VARSHITH COMPLE	Aging mechanism, performance measurement, testing, documentation, and result analysis.

7. Tools / Platforms / Software Used

Tool / Platform / Software Purpose

Linux / Ubuntu	Operating system and execution environment
C / GCC	Implementation and compilation
POSIX APIs	Process management and system programming
Linux System Calls	Process creation, control, synchronization, and priority management
Make file	Build and manage the project
Git / GitHub	Version control and project collaboration
GDB	Debugging and testing

8. Expected Outcome

A functional Linux-based Priority Scheduling System will be developed to create, manage, schedule, and monitor actual processes according to their priorities. The system will demonstrate pre-emptive and non pre-emptive scheduling and use aging to reduce starvation. It will generate scheduling results such as waiting time, turnaround time, response time, CPU utilization, and context switches, providing practical understanding of operating-system process scheduling and Linux/POSIX system programming.

9. GitHub Repository

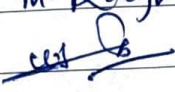
GitHub Repository Link: https://github.com/KLHB-2520080059/KLH-AIDS-2026-1-FOCUSFLOW

Repository Created: YES

☒ Yes ☐ No

Faculty Name and Signature

Faculty Name: M. Raghupathi

Signature: 

Remarks: _____

Faculty Use

Project Approved: Approved

☒ Yes ☐ No

Date: 24/2/2026

Faculty Signature: 